

PPA-6B — SCFSG Runtime, Adjacency Engine & Emulation Framework

Title

Runtime Systems and Emulation Methods for Symmetry-Constrained Finite-State Processors

Field of the Invention

The invention relates to runtime systems, schedulers, compilers, and emulation frameworks for novel processor architectures, particularly those composed of symmetry-constrained finite-state gates.

Background

Processors composed of finite-state elements with constrained transitions require execution environments distinct from conventional operating systems and compilers. In particular, when computation involves **dynamic adjacency**, **global constraint reconciliation**, or **block-level synchronization**, software support is essential.

There is a need for a runtime system that coordinates execution, manages adjacency relationships, and enables emulation of such processors on existing hardware.

Summary

This invention discloses a **runtime and emulation framework** for SCFSG-based processors. The runtime orchestrates execution of operator sequences, synchronizes microinstruction blocks, manages adjacency or neighborhood relationships among SCFSG elements, and enforces global symmetry or consistency constraints.

The framework enables:

- software emulation of SCFSG CPUs on conventional hardware,
 - hybrid execution (hardware SCFSG cores + software control),
 - compilation of higher-level descriptions into operator sequences.
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Detailed Description

1. Execution Runtime

The runtime system:

- schedules operator streams for SCFSG cores,
 - advances execution in microsteps or blocks,
 - collects state outputs at synchronization points.
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2. Adjacency and Neighborhood Management

For systems where SCFSG elements interact:

- adjacency relationships are stored and updated in software,
- rewiring or reassignment occurs at block boundaries,
- rules ensure global symmetry or constraint consistency.

Adjacency is treated as mutable state, not fixed wiring.

3. Emulation Framework

In a preferred embodiment:

- each SCFSG is emulated as a software object with an 8-bit state,
- transitions are computed via lookup tables,
- large arrays of SCFSGs are simulated in parallel.

This enables rapid prototyping and validation prior to hardware deployment.

4. Compilation and Mapping

The runtime may include:

- compilers translating high-level descriptions into operator sequences,
 - tools for mapping problems onto SCFSG arrays,
 - debugging and visualization tools showing state evolution.
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5. Hybrid Systems

The runtime supports hybrid configurations in which:

- SCFSG hardware executes state transitions,
 - a host processor manages adjacency, scheduling, and I/O.
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Advantages

- Enables practical use of SCFSG CPUs
 - Decouples hardware simplicity from global complexity
 - Facilitates experimentation and commercialization
 - Supports gradual transition from emulation to silicon
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Provisional Statement

This application claims priority for runtime systems, adjacency engines, and emulation frameworks for processors built from symmetry-constrained finite-state gates. It is independently enableable and may be practiced without specialized hardware.